

What makes a Copilot+ PC?

How do you know if a laptop is truly “AI-ready”? Microsoft has set clear minimum hardware requirements. To be considered a Copilot+ PC, a system must have at least 16 GB of RAM, a 256 GB SSD, and – most critically – an on-board NPU capable of 40 TOPS (trillions of operations per second). In simpler terms, the laptop needs a built-in AI brain that can perform 40 trillion computations each second for AI tasks. NPUs are specialized processors separate from the CPU and GPU, designed specifically for the heavy math of machine learning algorithms. They handle things like image recognition, language models, and other AI workloads far more efficiently than a traditional processor or graphics card.



Why insist on an NPU instead of using a powerful GPU for AI? The answer comes down to efficiency – especially important in a battery-powered laptop. GPUs can indeed crunch AI models (for instance, an NVIDIA RTX 4050 laptop GPU can theoretically deliver ~47–194 TOPS, and a high-end RTX 4090 up to 800 TOPS). However, those numbers don't tell the whole story. A GPU achieving those TOPS might drain your battery in an hour or two, running hot and loud. In contrast, NPUs are purpose-built for AI, as they deliver great performance per watt, enabling all-day battery life even while running AI tasks. Microsoft's Copilot+ standard deliberately excludes GPUs from the TOPS requirement – no matter how fast your graphics card is, it won't qualify unless there's an NPU in the system. The idea is that AI features should be available on the go, without tethering you to a power outlet or requiring cloud compute.

In practical terms, a Copilot+ laptop can do things like transcribe meetings, generate images or summaries, and translate languages in real time, entirely on-device. This not only makes these features faster and more private (since your data isn't constantly sent to the internet), but it also means the PC remains responsive while multitasking. Early users of AI PCs report that using the built-in Windows Copilot (which may even have a dedicated keyboard key on these models) feels like having a smart assistant baked right into the computer's DNA. If that

prospect is exciting, then an AI-PC might be worth prioritizing in your next purchase.

AI-ready chips: AMD, Intel, and Qualcomm

In late 2024, only a handful of chips met the Copilot+ criteria – notably Qualcomm's Snapdragon X series, which was the first to hit the required 40+ TOPS mark with its dedicated Hexagon AI engine (around 45 TOPS). Those Snapdragon X Elite laptops (built on an ARM architecture) essentially kicked off the AI-PC era, and brands like HP, Lenovo, Acer, and Asus rolled out ultraportables built on that platform. However, by 2025 the x86 camp has caught up.

AMD and Intel have both launched laptop processors with beefy NPUs to join the party. AMD's Ryzen AI 300 series chips, based on the latest Zen 5 cores, come with NPUs delivering around 50 TOPS of AI performance, making even a mid-range Ryzen AI 5 330 fully Copilot+ compliant. This is a huge jump from AMD's first-gen “Ryzen AI” (in the 2023 Ryzen 7040 chips) which had only ~4 TOPS – in two years AMD went from dabbling in AI to meeting Microsoft's high bar for AI PCs. On the Intel side, the newest Intel Core Ultra (14th/15th Gen “Series 2”) mobile processors – codenamed Lunar Lake for the 200V series – include NPUs roughly in the 45–50 TOPS range as well.

What this means is that as of late 2025, you can find AI-ready laptops across all major processor brands:

- **Qualcomm Snapdragon X (e.g. Snapdragon X Elite):** 12-core Arm SoC with a 45 TOPS NPU – found in designs like the HP OmniBook X 14 and Acer Swift 14 AI, focused on sleek fanless ultrabooks.
- **Intel Core Ultra 200 series (e.g. Core Ultra 7 2xx chips):** x86 processors (Lunar Lake architecture) with integrated NPUs (~48 TOPS) – powering models such as the latest Dell XPS 13 Copilot edition and Surface Pro Copilot+ variants. These bring AI capabilities to mainstream business and ultrabook segments with Windows on x86.
- **AMD Ryzen AI 300 series (e.g. Ryzen AI 7350 or Ryzen AI 5330):** Zen 5-based APUs with up to 50 TOPS NPUs – appearing in devices like certain Lenovo Yoga Slim 7 and ThinkPad T14s Gen 5 models for those who prefer AMD's balance of strong CPU/GPU plus AI acceleration.

It's worth noting that all Copilot+ certified machines so far are laptops – desktop PCs haven't gotten NPUs yet in late 2025, and even high-end gaming laptops with big GPUs don't count unless they have the requisite NPU on board. The focus has been on mobile PCs because that's where efficient on-device AI makes the biggest difference (in terms of battery life and portability). As competition heats up, expect AI capability to trickle down to even more affordable laptops in 2026.

Buy now or wait? Deciding on a Copilot+ PC

With any new technology, a key question for buyers is timing. Should you insist on a Copilot+-ready laptop right now, or can